Лабораторная работа №2

Тема: Настройка протокола STP (IEEE 802.1D)

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета.

Из проекта Menzhulin-lab2-template была скопирована сеть (рис. 1).

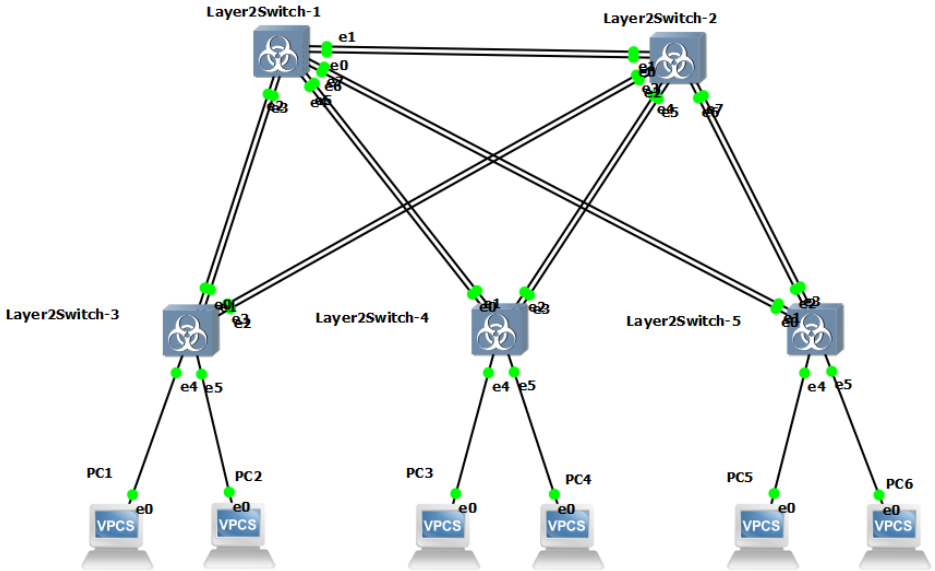

Рис. 1. Топология сети.

В качестве корневого коммутатора для сети был выбран Layer2Switch-1

Настройка адресов, масок и портов.

Hostname	IP-address	Subnet Mask	Gateway
Layer2Switch-1	192.168.10.1	255.255.255.0	
Layer2Switch-2	192.168.10.2	255.255.255.0	
Layer2Switch-3	192.168.10.3	255.255.255.0	
Layer2Switch-4	192.168.10.4	255.255.255.0	
PC1	192.168.10.11	255.255.255.0	192.168.10.254
PC2	192.168.10.12	255.255.255.0	192.168.10.254
PC3	192.168.10.21	255.255.255.0	192.168.10.254
PC4	192.168.10.22	255.255.255.0	192.168.10.254
PC5	192.168.10.31	255.255.255.0	192.168.10.254

PC6	192.168.10.32	255.255.255.0	192.168.10.254

Настройка коммутатора Layer2Switch-1:

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interface vl
vIOS-L2-01(config)#interface vlan1
vIOS-L2-01(config-if)#
*Apr 26 08:56:35.647: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Vlan1, changed state to down
*Apr 26 08:56:36.442: %PLATFORM-5-SIGNATURE_VERIFIED: Image
'flash0:/vios_l2-adventerprisek9-m' passed code signing verification
vIOS-L2-01(config-if)#ip address 192.168.10.1 255.255.255.0
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
vIOS-L2-01#
*Apr 26 08:57:05.429: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write mem
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5307 bytes to 2047 bytes[OK]
vIOS-L2-01#
*Apr 26 08:57:23.804: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 08:57:24.577: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

Настройка коммутатора Layer2Switch-2:

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interf
vIOS-L2-01(config)#interface vl
vIOS-L2-01(config)#interface vlan1
vIOS-L2-01(config-if)#
*Apr 26 09:05:36.920: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Vlan1, changed state to down
vIOS-L2-01(config-if)#ip add
vIOS-L2-01(config-if)#ip address 192.168.10.2 255.255.255.0
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
vIOS-L2-01#
*Apr 26 09:05:56.757: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write mem
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5307 bytes to 2048 bytes[OK]
vIOS-L2-01#
*Apr 26 09:06:06.458: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 09:06:07.232: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

Настройка коммутатора Layer2Switch-3:

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#inter
vIOS-L2-01(config)#interface vlan1
vIOS-L2-01(config-if)#
*Apr 26 09:08:12.248: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Vlan1, changed state to down
vIOS-L2-01(config-if)#ip address 192.168.10.3 255.255.255.0
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
vIOS-L2-01#
*Apr 26 09:08:32.090: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5109 bytes to 1978 bytes[OK]
vIOS-L2-01#
*Apr 26 09:08:48.159: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 09:08:48.913: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

Настройка коммутатора Layer2Switch-4:

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interface vlan1
vIOS-L2-01(config-if)#ip ad
*Apr 26 09:09:53.697: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Vlan1, changed state to down
vIOS-L2-01(config-if)#ip address 192.168.10.4 255.255.255.0
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
vIOS-L2-01#
*Apr 26 09:10:15.376: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5109 bytes to 1976 bytes[OK]
vIOS-L2-01#
*Apr 26 09:10:24.549: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 09:10:25.309: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

Настройка коммутатора Layer2Switch-5:

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interface vlan1
vIOS-L2-01(config-if)#ip a
*Apr 26 09:11:09.147: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Vlan1, changed state to down
vIOS-L2-01(config-if)#ip address 192.168.10.5 255.255.255.0
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
vIOS-L2-01#
```

```
*Apr 26 09:11:36.556: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5109 bytes to 1976 bytes[OK]
vIOS-L2-01#
*Apr 26 09:11:46.710: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 09:11:47.465: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

Настройка PC1:

```
PC1> ip 192.168.10.11 192.168.10.254 24
Checking for duplicate address...
PC1 : 192.168.10.11 255.255.255.0 gateway 192.168.10.254
```

Настройка PC2:

```
PC2> ip 192.168.10.12 192.168.10.254 24
Checking for duplicate address...
PC2 : 192.168.10.12 255.255.255.0 gateway 192.168.10.254
```

Настройка PC3:

```
PC3> ip 192.168.10.21 192.168.10.254 24
Checking for duplicate address...
PC3 : 192.168.10.21 255.255.255.0 gateway 192.168.10.254
```

Настройка PC4:

```
PC4> ip 192.168.10.22 192.168.10.254 24
Checking for duplicate address...
PC4 : 192.168.10.22 255.255.255.0 gateway 192.168.10.254
```

Настройка PC5:

```
PC5> ip 192.168.10.31 192.168.10.254 24
Checking for duplicate address...
PC5 : 192.168.10.31 255.255.255.0 gateway 192.168.10.254
```

Настройка PC6:

```
PC6> ip 192.168.10.32 192.168.10.254 24
Checking for duplicate address...
PC6 : 192.168.10.32 255.255.255.0 gateway 192.168.10.254
```

Проверка работоспособности:

С PC1:

PC1> ping 192.168.10.12

PC1> ping 192.168.10.21

PC1> ping 192.168.10.22

PC1> ping 192.168.10.31

PC1> ping 192.168.10.32

Результат

```
PC1> ping 192.168.10.12
```

```
84 bytes from 192.168.10.12 icmp_seq=1 ttl=64 time=1.213 ms
84 bytes from 192.168.10.12 icmp_seq=2 ttl=64 time=8.842 ms
84 bytes from 192.168.10.12 icmp_seq=3 ttl=64 time=1.954 ms
84 bytes from 192.168.10.12 icmp_seq=4 ttl=64 time=0.731 ms
84 bytes from 192.168.10.12 icmp_seq=5 ttl=64 time=3.246 ms

PC1> ping 192.168.10.21

84 bytes from 192.168.10.21 icmp_seq=1 ttl=64 time=9.079 ms
84 bytes from 192.168.10.21 icmp_seq=2 ttl=64 time=12.050 ms
84 bytes from 192.168.10.21 icmp_seq=3 ttl=64 time=1.812 ms
84 bytes from 192.168.10.21 icmp_seq=4 ttl=64 time=4.966 ms
84 bytes from 192.168.10.21 icmp_seq=5 ttl=64 time=7.969 ms

PC1> ping 192.168.10.22

84 bytes from 192.168.10.22 icmp_seq=1 ttl=64 time=9.949 ms
84 bytes from 192.168.10.22 icmp_seq=2 ttl=64 time=6.851 ms
84 bytes from 192.168.10.22 icmp_seq=3 ttl=64 time=4.612 ms
84 bytes from 192.168.10.22 icmp_seq=4 ttl=64 time=5.582 ms
84 bytes from 192.168.10.22 icmp_seq=5 ttl=64 time=8.561 ms

PC1> ping 192.168.10.31

84 bytes from 192.168.10.31 icmp_seq=1 ttl=64 time=6.473 ms
84 bytes from 192.168.10.31 icmp_seq=2 ttl=64 time=5.639 ms
84 bytes from 192.168.10.31 icmp_seq=3 ttl=64 time=12.360 ms
84 bytes from 192.168.10.31 icmp_seq=4 ttl=64 time=7.279 ms
84 bytes from 192.168.10.31 icmp_seq=5 ttl=64 time=7.708 ms

PC1> ping 192.168.10.32

84 bytes from 192.168.10.32 icmp_seq=1 ttl=64 time=6.264 ms
84 bytes from 192.168.10.32 icmp_seq=2 ttl=64 time=7.392 ms
84 bytes from 192.168.10.32 icmp_seq=3 ttl=64 time=8.890 ms
84 bytes from 192.168.10.32 icmp_seq=4 ttl=64 time=7.216 ms
84 bytes from 192.168.10.32 icmp_seq=5 ttl=64 time=6.730 ms
```

С PC1 успешно проходит сигнал на все остальные PC.

Проверка конфигураций STP на коммутаторах:

Layer2Switch-1:

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c67.da85.0000
             Cost        4
             Port        7 (GigabitEthernet1/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c9e.8e43.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  300 sec
```

```
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Root FWD 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

Layer2Switch-2:

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c67.da85.0000
             Cost        4
             Port        7 (GigabitEthernet1/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0ccb.cae5.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
Gi0/0               Altn BLK 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Root FWD 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

Layer2Switch-3:

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c67.da85.0000
             Cost        8
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cbc.9936.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  300 sec
```

```
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

Layer2Switch-4:

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c67.da85.0000
             Cost        8
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0ccf.fde1.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

Layer2Switch-5:

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c67.da85.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c67.da85.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
```

```
Gi1/1               Desg FWD 4         128.6    Shr
```

Из полученных конфигураций STP видно, что сейчас Layer2Switch-5 является корневым (This bridge is the root), т.к. у него самый малый MAC-адрес из всех.

Установка более низкого приоритета у Layer2Switch-1.

На Layer2Switch-1:

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spann
vIOS-L2-01(config)#spanning-tree vlan 1 pr
vIOS-L2-01(config)#spanning-tree vlan 1 priority 0
vIOS-L2-01(config)#exit
vIOS-L2-01#wr
*Apr 26 09:30:27.381: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write me
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5341 bytes to 2069 bytes[OK]
vIOS-L2-01#
*Apr 26 09:30:36.866: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 09:30:37.639: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

Проверка:

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c9e.8e43.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    1      (priority 0 sys-id-ext 1)
             Address     0c9e.8e43.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  15  sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

```
Gi1/2                Desg FWD 4         128.7    Shr
Gi1/3                Desg FWD 4         128.8    Shr
Gi2/0                Desg FWD 4         128.9    Shr
```

Все порты назначены и коммутатор имеет приоритет 1.

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

Повторная проверка доступности VPCS:

```
PC1> ping 192.168.10.12

84 bytes from 192.168.10.12 icmp_seq=1 ttl=64 time=7.158 ms
84 bytes from 192.168.10.12 icmp_seq=2 ttl=64 time=6.500 ms
84 bytes from 192.168.10.12 icmp_seq=3 ttl=64 time=3.879 ms
84 bytes from 192.168.10.12 icmp_seq=4 ttl=64 time=2.437 ms
84 bytes from 192.168.10.12 icmp_seq=5 ttl=64 time=0.965 ms

PC1> ping 192.168.10.21

84 bytes from 192.168.10.21 icmp_seq=1 ttl=64 time=4.762 ms
84 bytes from 192.168.10.21 icmp_seq=2 ttl=64 time=4.951 ms
84 bytes from 192.168.10.21 icmp_seq=3 ttl=64 time=4.071 ms
84 bytes from 192.168.10.21 icmp_seq=4 ttl=64 time=2.160 ms
84 bytes from 192.168.10.21 icmp_seq=5 ttl=64 time=7.762 ms

PC1> ping 192.168.10.22

84 bytes from 192.168.10.22 icmp_seq=1 ttl=64 time=6.906 ms
84 bytes from 192.168.10.22 icmp_seq=2 ttl=64 time=5.120 ms
84 bytes from 192.168.10.22 icmp_seq=3 ttl=64 time=1.767 ms
84 bytes from 192.168.10.22 icmp_seq=4 ttl=64 time=6.689 ms
84 bytes from 192.168.10.22 icmp_seq=5 ttl=64 time=7.890 ms

PC1> ping 192.168.10.31

84 bytes from 192.168.10.31 icmp_seq=1 ttl=64 time=7.140 ms
84 bytes from 192.168.10.31 icmp_seq=2 ttl=64 time=1.689 ms
84 bytes from 192.168.10.31 icmp_seq=3 ttl=64 time=13.172 ms
84 bytes from 192.168.10.31 icmp_seq=4 ttl=64 time=2.958 ms
84 bytes from 192.168.10.31 icmp_seq=5 ttl=64 time=7.537 ms

PC1> ping 192.168.10.32

84 bytes from 192.168.10.32 icmp_seq=1 ttl=64 time=11.204 ms
84 bytes from 192.168.10.32 icmp_seq=2 ttl=64 time=5.628 ms
84 bytes from 192.168.10.32 icmp_seq=3 ttl=64 time=6.906 ms
84 bytes from 192.168.10.32 icmp_seq=4 ttl=64 time=7.004 ms
84 bytes from 192.168.10.32 icmp_seq=5 ttl=64 time=7.284 ms
```

Аналогичные результаты на других компьютерах. Из чего можно сделать вывод, что сеть работает.

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

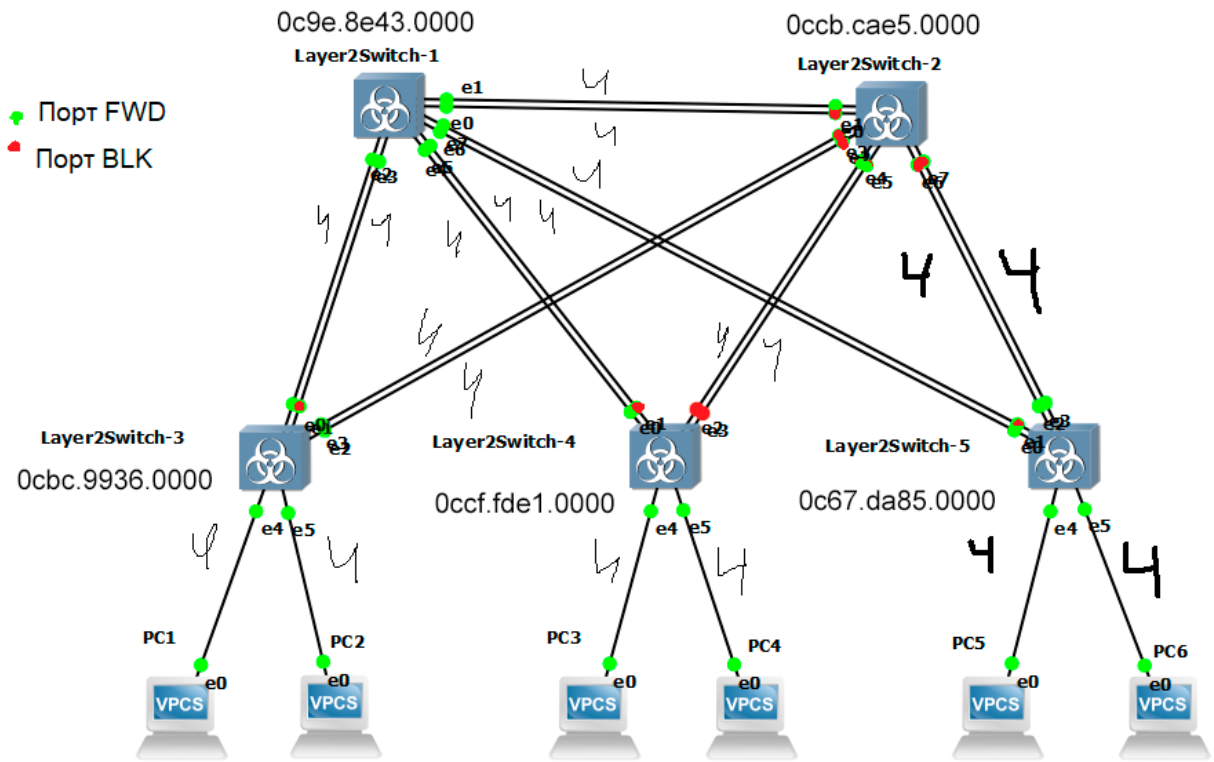

Рис. 2. Топология сети, BID и режимы работы портов

4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

Для анализа трафика в wireshark применён фильтр “stp”. Из захвата трафика между портами e0 и e0 Layer2Switch-1 и Layer2Switch-2 (рис.3) видно, как они пересылают друг другу сообщения

stp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 0/1/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
2	0.212166	0c:cb:ca:e5:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/100/0c:cb:ca:e5:00:00 Cost = 0 Port = 0x8001
3	0.217143	0c:cb:ca:e5:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/200/0c:cb:ca:e5:00:00 Cost = 0 Port = 0x8001
4	0.218503	0c:cb:ca:e5:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/300/0c:cb:ca:e5:00:00 Cost = 0 Port = 0x8001
5	0.999940	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 0/1/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
6	1.072455	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/100/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
7	1.075448	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/200/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
8	1.164409	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/300/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
9	1.999778	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 0/1/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
10	2.211829	0c:cb:ca:e5:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/100/0c:cb:ca:e5:00:00 Cost = 0 Port = 0x8001
11	2.216860	0c:cb:ca:e5:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/200/0c:cb:ca:e5:00:00 Cost = 0 Port = 0x8001
12	2.218154	0c:cb:ca:e5:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/300/0c:cb:ca:e5:00:00 Cost = 0 Port = 0x8001
13	2.999632	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 0/1/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
14	3.072168	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/100/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
15	3.075149	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/200/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001
16	3.164104	0c:9e:8e:43:00:00	Nearest-Customer-Br…	STP	60	Conf. Root = 32768/300/0c:9e:8e:43:00:00 Cost = 0 Port = 0x8001

Рис. 3. Передача сообщений между портами e0 Layer2Switch-1 (root) и e0 Layer2Switch-2.

MAC-адрес Layer2Switch-1 e0: 0c:9e:8e:43:00:00
И он имеет: Bridge Priority: 0 и Root Bridge Priority: 0

MAC-адрес Layer2Switch-2 e0: 0c:cb:ca:e5:00:00
И он имеет: Bridge Priority: 32768 и Root Bridge Priority: 32768

Оба имеют стоимость 0 (Root Path Cost: 0) и вещают на широковещательном адресе (Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00))

Аналогично и для остальных линках. Все имеют стоимость 0, вещают на широковещательном адресах, и имеют приоритет в 32768, если он не. корневой и у всех них Hello Time: 2.

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл.

У Layer2Switch-3 изменю стоимость e0.
Смена

На Layer2Switch-1 повышаем приоритет до 4096.

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spann
vIOS-L2-01(config)#spanning-tree vl
vIOS-L2-01(config)#spanning-tree vlan 1 pr
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4
% Bridge Priority must be in increments of 4096.
% Allowed values are:
  0     4096  8192  12288 16384 20480 24576 28672
  32768 36864 40960 45056 49152 53248 57344 61440
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#exit
```

```
vIOS-L2-01#w
*Apr 26 10:32:59.503: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write ,e
vIOS-L2-01#write me
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5344 bytes to 2075 bytes[OK]
vIOS-L2-01#
*Apr 26 10:33:10.812: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 10:33:11.583: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

На Layer2Switch-1 меняем приоритет на 0.

```
vIOS-L2-01>enable
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spann
vIOS-L2-01(config)#spanning-tree vl
vIOS-L2-01(config)#spanning-tree vlan 1
vIOS-L2-01(config)#spanning-tree vlan 1 pri
vIOS-L2-01(config)#spanning-tree vlan 1 priority 0
vIOS-L2-01(config)#exit
vIOS-L2-01#
*Apr 26 10:34:16.197: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#write mem
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5143 bytes to 1997 bytes[OK]
vIOS-L2-01#
*Apr 26 10:34:31.496: %GRUB-5-CONFIG_WRITING: GRUB configuration is
being updated on disk. Please wait...
*Apr 26 10:34:32.251: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was
written to disk successfully.
```

После

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0cbc.9936.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    1      (priority 0 sys-id-ext 1)
             Address     0cbc.9936.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------
------------
```

```
Gi0/0                Desg FWD 4          128.1     Shr
Gi0/1                Desg FWD 4          128.2     Shr
Gi0/2                Desg FWD 4          128.3     Shr
Gi0/3                Desg FWD 4          128.4     Shr
Gi1/0                Desg FWD 4          128.5     Shr
Gi1/1                Desg FWD 4          128.6     Shr
```

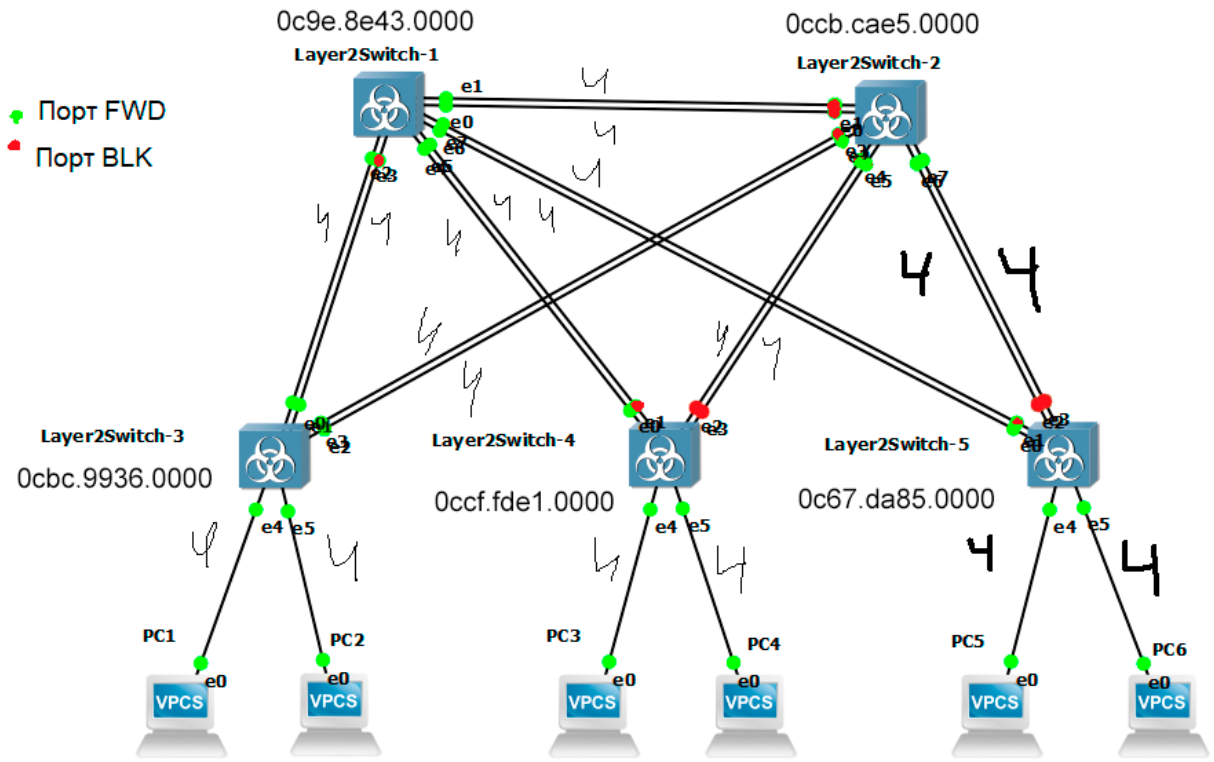

Рис. 4. Конфигурация сети, после изменения стоимости e0 у SW-3.